

Calabi–Yau quantum groups and 6d holomorphic Chern–Simons

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Abstract

Volume I of the modular Koszul duality programme introduces E_1 -chiral quantum groups as abstract algebraic-geometric objects: ordered bar complexes with spectral R -matrices, Drinfeld coproducts, and Yang–Baxter equations on configuration spaces of curves. This paper constructs them concretely from Calabi–Yau geometry.

The source is the holomorphic Chern–Simons hierarchy. At the top ($d = 6$): holomorphic Chern–Simons theory on a CY_3 -fold Y . A curve $C \subset Y$ of codimension 2 carries a defect algebra: the normal bundle $N_{C/Y}$ supplies two spectral parameters, and the defect algebra is a chiral algebra on C with quantum group structure from the normal directions. For $Y = \mathbb{C}^3$ and $C = \mathbb{C}$, the defect algebra is $\mathcal{W}_{1+\infty}$ at $\Psi = -\sigma_2(h_1, h_2, h_3)$. The hierarchy $6 \rightarrow 5 \rightarrow 4 \rightarrow 3$ is codimension-2 restriction at each step; each restriction produces a chiral algebra from a gauge theory, and the E_n level decreases as the dimension drops.

The critical cohomological Hall algebra $\text{CoHA}(Q, W)$ of a quiver with CY_3 potential is the E_1 -primitive: its product is ordered Hall multiplication, its R -matrix comes from the Ext-quiver, and its coproduct is the Schiffmann–Vasserot vertex coproduct. For the Jordan quiver: $\text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$. The Drinfeld center recovers E_2 -braiding, closing the E_n operadic circle.

Quantum toroidal algebras $U_{q,t}(\widehat{\mathfrak{gl}}_1)$ carry the CY condition as the parameter constraint $q_1 q_2 q_3 = 1$. The structure function $G(x; q_1, q_2, q_3) = \prod_i (1 - q_i x) / \prod_i (1 - q_i^{-1} x)$ satisfies the CY reflection identity $G(x) \cdot G(1/x) = 1$ precisely when $q_1 q_2 q_3 = 1$.

Wall-crossing in the Bridgeland stability manifold is Maurer–Cartan gauge equivalence in the convolution algebra of Volume I: the Kontsevich–Soibelman wall-crossing formula is the MC gauge transformation $\Theta_\sigma \sim \Theta_{\sigma'}$. The shadow obstruction tower at $\kappa_{\text{ch}} = \Psi$ produces the perturbative constant-map Gromov–Witten free energies $F_g^{\text{GW}, \text{const}}(\mathbb{C}^3)$: for $\mathbb{C}^3$ (no compact curves), the shadow IS the full GW answer.

Contents

1	Introduction	2
1.1	CY quantum groups as concrete E_1 -chiral quantum groups	2
1.2	The holomorphic Chern–Simons hierarchy	3
1.3	The E_n operadic circle	3
1.4	Outline	3
2	The holomorphic Chern–Simons hierarchy	4
2.1	6d holomorphic Chern–Simons	4
2.2	Dimensional reduction: $6 \rightarrow 5 \rightarrow 4 \rightarrow 3$	4
2.3	The Ω -background and equivariant parameters	4

3	Codimension-2 defect algebras	5
3.1	The defect construction	5
3.2	$\mathbb{C}^3$: the $\mathcal{W}_{1+\infty}$ identification	5
3.3	Defect algebras for general CY_3	6
4	The CoHA as E_1-primitive	7
4.1	Critical cohomological Hall algebras	7
4.2	The CoHA carries E_1 -chiral bialgebra data	7
4.3	The R -matrix and the Drinfeld coproduct	8
5	Quantum toroidal algebras and the CY condition	8
5.1	The CY parameter constraint	8
5.2	The Miki automorphism	9
5.3	The DDCA–toroidal bridge	9
6	The Drinfeld center closes the E_n circle	10
7	Shadow = $\text{GW}(\mathbb{C}^3)$ and wall-crossing	10
7.1	The shadow tower as Gromov–Witten theory	10
7.2	Wall-crossing as MC gauge equivalence	11
8	The CY_4 obstruction and the p_1-twisted double current algebra	11
8.1	The obstruction in $\pi_4(BU)$	12
8.2	The Drinfeld double presentation	12
8.3	The RTT relation through $O(\hbar^2)$	12
8.4	Seven-dimensional holomorphic Chern–Simons	13
8.5	Comparison with Maulik–Okounkov	13
8.6	Unobstructed E_4 on $K3 \times K3$	13
9	Wave-14 reconception: codim-2 defect, CoHA, toroidal	14

1 Introduction

1.1 CY quantum groups as concrete E_1 -chiral quantum groups

Volume I constructs E_1 -chiral quantum groups as abstract algebraic-geometric objects on curves. An E_1 -chiral quantum group is a tuple $(A, \mu, \Delta_z, \varepsilon, \eta, S)$ satisfying five axioms (H1–H5): an E_1 -chiral algebra structure, a spectral coproduct, bialgebra compatibility, spectral coassociativity

$$(\Delta_w \otimes \text{id}) \circ \Delta_{z+w} = (\text{id} \otimes \Delta_z) \circ \Delta_w,$$

and the Hopf axiom at $z = 0$. The ordered bar $B^{\text{ord}}(A) = T^c(s^{-1}\bar{A})$ preserves the R -matrix; the symmetric bar $B^\Sigma(A)$ kills the Hopf structure via the lossy averaging map: in abelian and scalar families $\text{av}(r(z)) = \kappa_{\text{ch}}$, while for non-abelian affine Kac–Moody $\text{av}(r(z)) + \dim(\mathfrak{g})/2 = \kappa_{\text{ch}}$.

Neither Volume I nor Volume II constructs a single E_1 -chiral quantum group. This paper constructs them from two sources: the 6d holomorphic Chern–Simons hierarchy (which produces chiral algebras as defect algebras on codimension-2 curves) and the critical cohomological

Hall algebra (which provides the E_1 -primitive as an associative algebra graded by dimension vectors).

The E_1 stabilization theorem of the companion paper [?] proves that for $d \geq 3$, the CY chiral algebra is natively E_1 : the Schouten–Nijenhuis bracket vanishes and all noncommutative structure arises from quantization. The braided E_2 -structure is recovered through the Drinfeld center of the E_1 -monoidal representation category, not through the native E_d -framing (which produces only symmetric braiding for $d \geq 3$, since $\pi_1(\text{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$).

1.2 The holomorphic Chern–Simons hierarchy

The theories of this paper form a hierarchy in real dimension:

d	Theory	Curve output	E_n level
6	hCS on CY_3 Y	defect on $C \subset Y$	E_1
5	partially topological CS on $\mathbb{R} \times S$	$\mathbb{R} \times C$	E_1
4	hCS on $\mathbb{C}^2$ (CWY)	line operator	E_1
3	3d HT (Volume II)	boundary algebra	E_1

The hierarchy $6 \rightarrow 5 \rightarrow 4 \rightarrow 3$ is codimension-2 restriction at each step. Each restriction produces a chiral algebra from a gauge theory. The resulting chiral algebras are E_1 -chiral quantum groups in the sense of Volume I, with R -matrices from the normal directions and coproducts from the spectral parameters.

1.3 The E_n operadic circle

The E_n hierarchy organizes the entire programme:

$$E_3(\text{bulk}) \rightarrow E_2(\text{boundary chiral}) \rightarrow E_1(\text{bar/QG}) \rightarrow E_2(\text{Drinfeld center}) \rightarrow E_3(\text{derived center}).$$

Each arrow has precise mathematical content: $E_3 \rightarrow E_2$ is restriction to a codimension-2 defect; $E_2 \rightarrow E_1$ is the ordered bar complex; $E_1 \rightarrow E_2$ is the Drinfeld center; $E_2 \rightarrow E_3$ is the higher Deligne conjecture. The circle closes for 3d HT theories with conformal vector at non-critical level (the topologization theorem, proved for affine KM); without conformal vector, the theory is stuck at $\text{SC}^{\text{ch}, \text{top}}$.

The CY quantum groups of this paper enter the circle at the E_1 vertex: the CoHA is the E_1 -primitive, and the Drinfeld center carries it to E_2 . The tetrahedron equation (3d analogue of YBE) fails for the Yang R -matrix at $O(\kappa_{\text{ch}}^2)$: the E_3 structure is genuinely nontrivial beyond E_2 .

1.4 Outline

Section 2 develops the hCS hierarchy from 6d to 3d with the codimension-2 defect mechanism. Section 3 constructs the codimension-2 defect algebras and proves the $\mathcal{W}_{1+\infty}$ identification for $\mathbb{C}^3$. Section 4 establishes the CoHA as the E_1 -primitive and develops the Hopf axioms. Section 5 treats quantum toroidal algebras and the CY condition $q_1 q_2 q_3 = 1$. Sections 6–7 develop the Drinfeld center and the shadow-equals-GW identification.

2 The holomorphic Chern–Simons hierarchy

2.1 6d holomorphic Chern–Simons

Let Y be a CY_3 -fold with holomorphic volume form $\Omega_Y \in H^0(Y, \omega_Y)$. The holomorphic Chern–Simons functional on Y (Witten, Costello) is

$$S_{\text{hCS}}(\mathcal{A}) = \int_Y \Omega_Y \wedge \text{tr} \left(\mathcal{A} \wedge \bar{\partial} \mathcal{A} + \frac{2}{3} \mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A} \right)$$

where $\mathcal{A} \in \Omega^{0,1}(Y, \mathfrak{g})$ is a $(0,1)$ -form valued in a Lie algebra $\mathfrak{g}$. The equations of motion $\bar{\partial} \mathcal{A} + \mathcal{A} \wedge \mathcal{A} = 0$ are the integrability condition for the holomorphic structure on the trivial G -bundle.

The theory is six-dimensional, holomorphic in all three complex directions. The BRST operator is $Q_{\text{BRST}} = \bar{\partial} + [\mathcal{A}, -]$, and the observables are the Dolbeault cohomology $H_{\bar{\partial}_{\mathcal{A}}}^{0,\bullet}(Y, \mathfrak{g})$.

Remark 2.1 (Relation to physical dimensions). The 6d hCS theory is not a physical quantum field theory in the Wightman sense: the kinetic term is first-order, the path integral is one-loop exact, and the partition function is a holomorphic invariant. It is the holomorphic twist of the 6d $(2,0)$ theory on a CY_3 -fold, and its observables produce chiral algebras by restriction to codimension-2 defects. The physical content is in the defect algebras, not in the bulk theory.

2.2 Dimensional reduction: $6 \rightarrow 5 \rightarrow 4 \rightarrow 3$

The hierarchy proceeds by codimension-2 restriction at each step.

$6 \rightarrow 5$: for a CY_3 -fold of the form $Y = \mathbb{R} \times S$ (where S is a CY_2 surface), the 6d hCS theory reduces to 5d partially topological Chern–Simons on $\mathbb{R} \times S$. The topological direction $\mathbb{R}$ carries the E_1 ordering; the holomorphic surface S carries the chiral data. Restriction to $\mathbb{R} \times C$ (where $C \subset S$ is a curve) produces the ordered bar complex with one spectral parameter.

$5 \rightarrow 4$: for $S = \mathbb{C}^2$, the 5d theory reduces to 4d holomorphic Chern–Simons on $\mathbb{C}^2$: the Costello–Witten–Yamazaki theory. Line operators on $\mathbb{C} \subset \mathbb{C}^2$ produce Yangians from holomorphic gauge fields. The R -matrix comes from the normal direction to the line.

$4 \rightarrow 3$: restricting to a further codimension-2 locus produces the 3d holomorphic-topological theories of Volume II. The boundary algebra is a chiral algebra on the codimension-2 curve, and the $\text{SC}^{\text{ch}, \text{top}}$ pair $(C_{\text{ch}}^{\bullet}(A, A), A)$ is the boundary–bulk system.

Proposition 2.2 (Dimensional consistency). *At each step of the hierarchy, the E_n level of the chiral algebra on the defect curve matches the ambient dimension: 6d produces E_1 -chiral (two normal directions give two spectral parameters, but the CY condition reduces to E_1); 5d produces E_1 -chiral with one spectral parameter; 4d produces E_1 -chiral (Yangian); 3d produces E_1 -chiral (boundary algebra of Volume II). The E_2 -braiding is recovered at each level through the Drinfeld center.*

2.3 The Ω -background and equivariant parameters

For the toric CY_3 -fold $Y = \mathbb{C}^3$, the three coordinate directions carry equivariant parameters $h_1, h_2, h_3 \in \mathbb{C}$ subject to the CY condition $h_1 + h_2 + h_3 = 0$. In the multiplicative parametrization $q_i = e^{h_i}$, the CY condition reads $q_1 q_2 q_3 = 1$. The identification $q_1 = q$, $q_2 = t^{-1}$, $q_3 = t/q$ recovers the standard (q, t) -parametrization of the quantum toroidal algebra.

The Ω -background parameters enter the defect algebra through the equivariant Chern–Simons functional: the gauge field on Y decomposes along the normal bundle to a defect curve $C \subset Y$, and the equivariant parameters weight the normal modes. For $C = \mathbb{C}_z$ (the first coordinate axis) inside $Y = \mathbb{C}^3$, the normal bundle is $N_{C/Y} = \mathcal{O}_C^{\oplus 2}$ with fiber coordinates (w_1, w_2) carrying equivariant weights (h_2, h_3) . The resulting defect algebra on C has OPE poles involving both normal parameters.

3 Codimension-2 defect algebras

3.1 The defect construction

A curve $C \subset Y$ in a CY_3 -fold carries a defect algebra: the 6d hCS theory restricted to a tubular neighborhood of C produces a 2d chiral algebra on C with additional structure from the normal bundle.

Construction 3.1 (Codimension-2 defect algebra). Let Y be a CY_3 -fold, $C \subset Y$ a smooth curve, and $N_{C/Y}$ the normal bundle of rank 2. The defect algebra $\mathcal{A}_C^{\text{def}}$ on C is obtained by:

- Step 1. Restriction.** Restrict the 6d hCS gauge field $\mathcal{A} \in \Omega^{0,1}(Y, \mathfrak{g})$ to the formal neighborhood $\hat{N}_{C/Y}$ of C in Y . The Taylor expansion along the normal fibers gives a collection of fields on C indexed by normal-direction monomials.
- Step 2. BRST reduction.** The BRST operator $Q = \bar{\partial} + [\mathcal{A}, -]$ restricts to the formal neighborhood. The Q -cohomology in the fiber directions produces the generating fields of the defect algebra on C .
- Step 3. OPE from normal modes.** The OPE on C is inherited from the 6d hCS action. The normal bundle $N_{C/Y}$ supplies two spectral parameters (from the two fiber coordinates), which enter the OPE as meromorphic functions of the insertion points on C .

Remark 3.2 (Two spectral parameters). The rank-2 normal bundle $N_{C/Y}$ gives two spectral parameters. This is the geometric origin of the two-parameter family of quantum toroidal algebras: the two normal directions to a curve inside a CY_3 -fold produce the (q, t) -deformation. For a CY_2 -fold (surface), the normal bundle has rank 1, producing one spectral parameter: the standard spectral parameter of Yangians and KZ equations.

3.2 $\mathbb{C}^3$: the $\mathcal{W}_{1+\infty}$ identification

Proposition 3.3 (Defect algebra of $\mathbb{C}^3$). *For $Y = \mathbb{C}^3$ with coordinates (z, w_1, w_2) and CY form $\Omega = dz \wedge dw_1 \wedge dw_2$, the codimension-2 defect algebra on $C = \mathbb{C}_z$ (the z -axis) is the $\mathcal{W}$ -algebra $\mathcal{W}_{1+\infty}$ at deformation parameter*

$$\Psi = -\sigma_2(h_1, h_2, h_3)$$

where σ_2 is the second elementary symmetric polynomial in the Ω -background parameters. The central charge is $c_{\mathcal{W}} = 1$ (the Sugawara value from the $\mathfrak{gl}_1$ current subalgebra).

Proof sketch. The gauge field $\mathcal{A} \in \Omega^{0,1}(\mathbb{C}^3, \mathfrak{gl}_N)$ expands along the normal fiber as $\mathcal{A}(z, w_1, w_2) = \sum_{a,b \geq 0} J_{a,b}(z) w_1^a w_2^b d\bar{z}$, where $J_{a,b}$ are fields on $C = \mathbb{C}_z$. The BRST cohomology in the (w_1, w_2) -directions retains one field at each bi-degree (a, b) with $a + b = s - 1$ for spin s : these are the

generators of $\mathcal{W}_{1+\infty}$. At spin 1: a single current $J(z)$ (the $\mathfrak{gl}_1$ current). At spin 2: the stress tensor $T(z)$ and the $U(1)$ current. The OPE is computed from the 6d hCS action by restricting to the formal neighborhood; the resulting algebra is $\mathcal{W}_{1+\infty}$ at $\Psi = -\sigma_2 = -(h_1h_2 + h_1h_3 + h_2h_3)$.

The CY condition $h_1 + h_2 + h_3 = 0$ gives $\sigma_2 = -(h_1^2 + h_2^2 + h_3^2)/2$, so $\Psi = (h_1^2 + h_2^2 + h_3^2)/2 > 0$ generically. The central charge of the $\mathcal{W}_{1+\infty}$ algebra at $N = 1$ (single $\mathfrak{gl}_1$ current) is computed from the Sugawara construction $T = (1/(2\Psi)) : J^2 :$ applied to the level- Ψ Heisenberg current $J(z)J(w) \sim \Psi/(z-w)^2$; this gives $c = 1$ for every nonzero Ψ , independent of the specific value. For N bosons the central charge is $c = N$; the 6d $\mathfrak{gl}_1$ gauge reduction produces $N = 1$. $\square$

Remark 3.4 (Level-prefixed r -matrix). The classical r -matrix of the $\widehat{\mathfrak{gl}}_1$ current subalgebra derived from the 6d reduction carries level $\Psi = -\sigma_2$, not a normalised level $k = 1$:

$$r^{\text{Heis}}(z) = \frac{\Psi}{z} = \frac{-\sigma_2(h_1, h_2, h_3)}{z}.$$

At $\Psi = 0$ the r -matrix vanishes and the Sugawara construction is singular; this is the degenerate locus sitting outside the non-critical Koszul locus. The averaging map gives $\text{av}(r(z)) = \Psi = \kappa_{\text{ch}}(H_\Psi)$. For the full $\mathcal{W}_{1+\infty}$ algebra, the r -matrix has additional components at higher spins, with the universal Miura coefficient $(\Psi - 1)/\Psi$ on the leading cross-term $J \otimes W_{s-1}$ at every spin $s \geq 2$ (Vol I `thm:miura-cross-universality`). The central charge $c = 1$ is the $N = 1$ rank of the underlying free boson and does not track Ψ ; Prochazka–Rapcak parametrise $\mathcal{W}_{1+\infty}$ equivalently by $(\lambda_1, \lambda_2, \lambda_3)$ with $\sum_i 1/\lambda_i = 0$: the identification with the 6d equivariant parameters is $\lambda_i^{-1} \propto h_i$ up to the homogeneity that makes Ψ projective in the h_i ; the $\Psi = -\sigma_2$ presentation used here fixes a normalisation consistent with Costello’s 6d action and with the `hcs.codim2.defect.ope` engine’s $J(z)J(w) \sim \Psi/(z-w)^2$ OPE.

3.3 Defect algebras for general CY₃

For a general CY₃-fold Y with a curve $C \subset Y$, the defect algebra depends on the normal bundle $N_{C/Y}$. The normal bundle is a rank-2 vector bundle on C ; its splitting type and degree control the defect algebra.

Example 3.5 (Conifold). For the resolved conifold $Y = \text{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \rightarrow \mathbb{P}^1)$ and $C = \mathbb{P}^1$ (the zero section): $N_{C/Y} = \mathcal{O}(-1)^{\oplus 2}$. The defect algebra on C has $\kappa_{\text{ch}} = 1$ (from the DT computation) and shadow class G (shadow depth 2). The negative degrees of the normal bundle constrain the spin content: only finitely many spins contribute, and the resulting algebra is a truncation of $\mathcal{W}_{1+\infty}$.

Example 3.6 (Local $\mathbb{P}^2$). For $Y = \text{Tot}(K_{\mathbb{P}^2} \rightarrow \mathbb{P}^2)$ and a line $C = \mathbb{P}^1 \subset \mathbb{P}^2$: $N_{C/Y} = \mathcal{O}(1) \oplus \mathcal{O}(-3)$. The defect algebra has $\kappa_{\text{ch}} = 3/2$ and shadow class M (infinite shadow depth). The mixed splitting type of the normal bundle introduces asymmetry between the two spectral parameters.

4 The CoHA as E_1 -primitive

4.1 Critical cohomological Hall algebras

For a quiver with potential (Q, W) arising from a toric CY_3 -fold, the critical CoHA is the $\mathbb{Z}$ -graded associative algebra

$$\mathrm{CoHA}(Q, W) = \bigoplus_{\mathbf{d} \in \mathbb{Z}_{\geq 0}^{Q_0}} H_*^{\mathrm{BM}}(\mathrm{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$$

where $\phi_{W_{\mathbf{d}}}$ is the vanishing cycle sheaf and the multiplication comes from Hall-algebra extension of representations. The grading by dimension vectors $\mathbf{d}$ is the CY analogue of the weight-space decomposition of a quantum group.

Theorem 4.1 (Schiffmann–Vasserot; Rapcak–Soibelman–Yang–Zhao). *For a toric CY_3 X without compact 4-cycles:*

$$\mathrm{CoHA}(Q_X, W_X) \simeq Y^+(\widehat{\mathfrak{g}}_{Q_X})$$

where $Y^+(\widehat{\mathfrak{g}}_{Q_X})$ is the positive half of the affine super Yangian attached to the toric quiver. For $\mathbb{C}^3$ (Jordan quiver, cubic potential): $\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$.

4.2 The CoHA carries E_1 -chiral bialgebra data

The CoHA is the natural E_1 -primitive of the CY quantum group. Its algebraic structure matches the E_1 -chiral bialgebra axiom system of Volume I; the chiral antipode does not lift (Remark 4.3), so the object is a chiral bialgebra in the sense of that volume, not a chiral Hopf algebra.

Proposition 4.2 (CoHA E_1 -chiral bialgebra data). *The critical CoHA $\mathrm{CoHA}(Q, W)$ of a toric CY_3 carries the data of an E_1 -chiral bialgebra, with a classical Yangian antipode that does not lift to a vertex-algebraic antipode:*

- (H1) E_1 -chiral algebra. *The Hall multiplication (extension of short exact sequences $0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0$ with sub before quotient) is an ordered (E_1) product. The ordering is physical: the dimension-vector filtration imposes a preferred direction.*
- (H2) Spectral coproduct. *The Schiffmann–Vasserot vertex coproduct Δ_z is an algebra homomorphism $\Delta_z: \mathrm{CoHA}(Q, W) \rightarrow \mathrm{CoHA}(Q, W) \otimes \mathrm{CoHA}(Q, W)((z))$ depending meromorphically on a spectral parameter z . At $z = 0$, it reduces to the deconcatenation coproduct on the ordered bar complex.*
- (H3) Bialgebra compatibility. *The coproduct Δ_z is a morphism of E_1 -chiral algebras: it intertwines the Hall multiplication with the tensor product multiplication.*
- (H4) Spectral coassociativity. *The coproduct satisfies $(\Delta_w \otimes \mathrm{id}) \circ \Delta_{z+w} = (\mathrm{id} \otimes \Delta_z) \circ \Delta_w$ with shifted parameters (not Δ_z composed with itself).*
- (H5) Classical antipode, non-lifting at the chiral level. *On the classical Yangian fibre, the antipode $S: \mathrm{CoHA}(Q, W) \rightarrow \mathrm{CoHA}(Q, W)^{\mathrm{op}}$ satisfies the standard Hopf identity at $z = 0$. For $\mathcal{W}_{1+\infty}$: $S(T(u)) = T(u)^{-1}$ in the transfer-matrix formalism. This classical*

antipode does not lift to a vertex-algebraic antipode on $\text{CoHA}(Q, W)$ (OPE obstruction $S(T)_{(3)}S(T) \neq c/2$ at generic Ψ , and the universal Hopf-axiom residual; see Remark 4.3 for the precise obstruction and Vol I Proposition ?? for the proved negative). The CoHA is therefore a chiral bialgebra, not a chiral Hopf algebra.

Remark 4.3 (Antipode non-lifting). The antipode $S(T(u)) = T(u)^{-1}$ is well-defined at the level of the transfer matrix (a generating function of Yangian generators). It does not lift to a vertex-algebraic antipode: two independent obstructions block the lift (OPE: the quartic pole $S(T)_{(3)}S(T) \neq c/2$ at generic Ψ ; Hopf: the $z \cdot J$ residual in the convolution $\mu \circ (S \otimes \text{id}) \circ \Delta_0$). The source of both obstructions is the Miura nonlinearity.

4.3 The R -matrix and the Drinfeld coproduct

The R -matrix of the CoHA arises from the Ext-quiver: the extension groups $\text{Ext}^1(V', V'')$ between representations determine the poles of the R -matrix, and the CY condition (self-duality of the Ext-quiver) constrains the R -matrix to satisfy the Yang–Baxter equation.

Proposition 4.4 (Universal Miura coefficient). *The Drinfeld coproduct Δ_z for $\mathcal{W}_{1+\infty}$ has the universal Miura coefficient $(\Psi - 1)/\Psi$ on the leading cross-term $J \otimes W_{s-1} + W_{s-1} \otimes J$ at every spin $s \geq 2$. At spin 2:*

$$\Delta_z(T) = T \otimes 1 + 1 \otimes T + \frac{\Psi - 1}{\Psi} J \otimes J + O(z^{-1}).$$

The coefficient $(\Psi - 1)/\Psi$ is structural: it arises from the Prochazka–Rapcak Miura factorization $T(u) = \prod_i (u + \Lambda_i)$ in three steps: (1) single- J sector gives $1/\Psi$; (2) Drinfeld shift adds $+1$; (3) lower sectors do not contribute. The coefficient persists at all spins $s \geq 2$ with 142 tests verifying spins 2–6.

Remark 4.5 (Coincidental agreement at $\Psi = 2$). At $\Psi = 2$: $(\Psi - 1)/\Psi = 1/2 = 1/\Psi$. The two expressions $(\Psi - 1)/\Psi$ and $1/\Psi$ agree at this single point, which is the self-dual level. The distinction is visible at $\Psi = 3$: $(\Psi - 1)/\Psi = 2/3 \neq 1/3 = 1/\Psi$. Verify at three or more parameter values to distinguish the two.

5 Quantum toroidal algebras and the CY condition

The quantum toroidal algebra $U_{q,t}(\widehat{\mathfrak{gl}}_1)$ is the two-parameter deformation of the double current algebra $\mathfrak{gl}_1 \otimes \mathbb{C}[u^{\pm 1}, v^{\pm 1}]$, carrying the full $\mathbb{Z} \times \mathbb{Z}$ bigrading from the two loop directions.

5.1 The CY parameter constraint

In the additive parametrization $q_i = e^{h_i}$, the three equivariant parameters of the Ω -background satisfy the CY condition:

$$h_1 + h_2 + h_3 = 0 \quad \text{equivalently} \quad q_1 q_2 q_3 = 1.$$

Two independent parameters survive. The standard (q, t) -parametrization identifies $q_1 = q$, $q_2 = t^{-1}$, $q_3 = t/q$.

Definition 5.1 (CY reflection identity). The *trigonometric structure function* of the quantum toroidal algebra is

$$G(x; q_1, q_2, q_3) = \frac{(1 - q_1x)(1 - q_2x)(1 - q_3x)}{(1 - q_1^{-1}x)(1 - q_2^{-1}x)(1 - q_3^{-1}x)}.$$

The *CY reflection identity* is

$$G(x) \cdot G(1/x) = 1.$$

Proposition 5.2 (CY reflection characterizes the CY condition). *The CY reflection identity $G(x) \cdot G(1/x) = 1$ holds if and only if $q_1q_2q_3 = 1$. Its failure measures the departure from Calabi–Yau: writing $q_1q_2q_3 = \omega$, the identity becomes $G(x) \cdot G(1/x) = \omega^2$, and the deviation $\omega^2 - 1$ is the anomaly.*

Proof. Direct computation. The numerator of $G(x) \cdot G(1/x)$ is $\prod_i (1 - q_i x)(1 - q_i/x) = \prod_i (1 - q_i x)(x - q_i)/x$. The denominator is $\prod_i (1 - q_i^{-1}x)(1 - q_i^{-1}/x) = \prod_i (1 - q_i^{-1}x)(x - q_i^{-1})/x$. The ratio simplifies to $\prod_i (q_i/q_i^{-1}) = \prod_i q_i^2 = (q_1q_2q_3)^2$, which equals 1 precisely when $q_1q_2q_3 = 1$. $\square$

5.2 The Miki automorphism

The Miki automorphism of $U_{q,t}(\widehat{\mathfrak{gl}}_1)$ is the cyclic permutation $(q_1, q_2, q_3) \mapsto (q_2, q_3, q_1)$, generating the S_3 -Weyl group of the CY torus. This automorphism is algebra-specific: it requires the concrete structure of the quantum toroidal algebra and does not follow from the E_3 -operad alone.

Remark 5.3 (The Miki automorphism is not operadic). The E_3 -operad acts on the derived center $Z_{\text{ch}}^{\text{der}}(A_{\mathcal{C}})$ through the higher Deligne conjecture. The Miki automorphism acts on the quantum toroidal algebra itself, permuting the three equivariant directions. These are different actions at different categorical levels: the E_3 -action is on the bulk algebra (the derived center), while the Miki automorphism is on the boundary algebra (the chiral algebra itself). The relationship between them is mediated by the E_n operadic circle, but the Miki automorphism is not a formal consequence of operadic data.

5.3 The DDCA–toroidal bridge

The deformed double current algebra $\text{DDCA}_k(\mathfrak{gl}_K) = U(\mathfrak{gl}_K \otimes \mathbb{C}[u, v])$ carries two spectral parameters from the two holomorphic directions of $\mathbb{C}^2$ in the 5d theory. The quantum toroidal algebra $U_{q,t}(\widehat{\mathfrak{gl}}_K)$ carries two deformation parameters from the two loop directions.

Conjecture 5.4 (DDCA–toroidal bridge). *Writing $q = e^{\hbar_1}$, $t = e^{\hbar_2}$, the DDCA is the rational $(\hbar_1, \hbar_2 \rightarrow 0)$ degeneration of the quantum toroidal algebra in the category of filtered algebras. The two one-parameter degenerations produce a commuting square:*

$$\begin{array}{ccc} U_{q,t}(\widehat{\mathfrak{gl}}_K) & \xrightarrow{t \rightarrow 1} & Y(\widehat{\mathfrak{gl}}_K) \otimes U(\mathbb{C}[v]) \\ \downarrow q \rightarrow 1 & & \downarrow \hbar_1 \rightarrow 0 \\ U(\mathbb{C}[u]) \otimes Y(\widehat{\mathfrak{gl}}_K) & \xrightarrow{\hbar_2 \rightarrow 0} & \text{DDCA}_k(\mathfrak{gl}_K) \end{array}$$

The diagonal $\hbar_1 = \hbar_2 \rightarrow 0$ recovers the affine Yangian $Y(\widehat{\mathfrak{gl}}_K)$ as the common degeneration. The modular characteristic is continuous across the bridge: $\kappa_{\text{ch}}(\text{DDCA}_k(\mathfrak{gl}_K)) = K(k+K)/2 = \lim_{\hbar_i \rightarrow 0} \kappa_{\text{ch}}(U_{q,t})$.

6 The Drinfeld center closes the E_n circle

The Drinfeld center $\mathcal{Z}(\text{Rep}^{E_1}(A))$ is the universal braided monoidal category receiving a central monoidal functor from $\text{Rep}^{E_1}(A)$. For the CoHA of a toric CY₃-fold:

Theorem 6.1 (Drinfeld center for CY₃ chiral algebras). *For A_C the E_1 -chiral algebra of a toric CY₃-fold X :*

$$\mathcal{Z}(\text{Rep}^{E_1}(A_C)) \simeq \text{Rep}^{E_2}(Z_{\text{ch}}^{\text{der}}(A_C))$$

where $Z_{\text{ch}}^{\text{der}}(A_C) = C_{\text{ch}}^{\bullet}(A_C, A_C)$ is the chiral derived center. The forgetful functor $U: \mathcal{Z}(\text{Rep}^{E_1}(A_C)) \rightarrow \text{Rep}^{E_1}(A_C)$ is the categorified averaging map.

Remark 6.2 (Naive center vs derived center). For $A = H_k$ (Heisenberg at level k): the naive center $Z(A) = \mathbb{C}$ has dimension 1; the derived center $Z_{\text{ch}}^{\text{der}}(A)$ has dimension 3, capturing Ext^1 and Ext^2 invisible to the commutant. The five invariants matching at six levels (conjectured for Heisenberg; proved end-to-end for $\mathbb{C}^3$ via the CoHA) confirm that the Drinfeld center of the E_1 -monoidal representation category is the correct categorification of the chiral derived center.

The passage $E_1 \rightarrow E_2$ via the Drinfeld center is the CY incarnation of the E_1 -to- E_2 lift of Volume I. The averaging map $\text{av}: \mathfrak{g}^{E_1} \rightarrow \mathfrak{g}^{\text{mod}}$ projects the ordered R -matrix to the scalar κ_{ch} ; the Drinfeld center is the categorical analogue, recovering the R -matrix data as the braiding on the center category.

Direct restriction from E_d to E_2 (for $d \geq 3$) gives only symmetric braiding: $\pi_1(\text{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$ for $d \geq 3$. The Drinfeld center is the sole source of non-trivial braiding for CY₃ quantum groups.

7 Shadow = GW($\mathbb{C}^3$) and wall-crossing

7.1 The shadow tower as Gromov–Witten theory

At $\kappa_{\text{ch}} = \Psi$ (the deformation parameter of $\mathcal{W}_{1+\infty}$ at $c = 1$), the shadow obstruction tower of Volume I produces the perturbative constant-map Gromov–Witten free energies $F_g^{\text{GW}, \text{const}}(\mathbb{C}^3)$: the shadow is the full GW answer for $\mathbb{C}^3$ (which has no compact curves, hence no worldsheet instantons).

Proposition 7.1 (Shadow = GW for $\mathbb{C}^3$). *The shadow coefficients S_r of the Virasoro algebra at the GW specialization $\kappa_{\text{ch}} = \Psi$ are the A_{∞} coproduct correction coefficients $\delta^{(r)}$: the shadow tower encodes the coproduct corrections of the chiral quantum group. On the DT side, the MacMahon function $M(q) = \prod_{n \geq 1} (1 - q^n)^{-n}$ encodes the partition function via the MNOP correspondence.*

The identification shadow = GW operates at three levels:

- (i) *Genus-1*: the shadow Eisenstein theorem $D_2(\mathcal{A}, s) = -24\kappa_{\text{ch}} \cdot \zeta(s)\zeta(s-1)$ (Volume I) gives the genus-1 DT partition function via the Borchers lift.

- (ii) *Virtual*: the shadow coefficient S_r at degree r matches the virtual Euler characteristic of the moduli of r -dimensional sheaves.
- (iii) *Motivic*: the full chiral quantum group structure (R -matrix, coproduct, RTT) should match the motivic DT invariants of Kontsevich–Soibelman.

7.2 Wall-crossing as MC gauge equivalence

For a toric CY_3 with fan Σ , Bridgeland stability conditions partition the space of central charges into chambers. Each chamber σ determines a quiver presentation (Q_σ, W_σ) and a CoHA $\text{CoHA}(Q_\sigma, W_\sigma)$.

Conjecture 7.2 (Wall-crossing = MC gauge equivalence). *Wall-crossing between stability chambers σ and σ' is Maurer–Cartan gauge equivalence in the modular convolution algebra: $\Theta_\sigma \sim \Theta_{\sigma'}$. The Kontsevich–Soibelman wall-crossing formula is the MC gauge transformation. The global chiral algebra is the homotopy colimit over the chamber decomposition: $A_C = \text{hocolim}_\sigma A_{Q_\sigma}$.*

For $\mathbb{C}^3$ (single chamber): the conjecture is vacuous. For the resolved conifold (two chambers): the wall-crossing formula is the flop relation on DT partition functions (Bryan–Steinberg), and the MC gauge equivalence is the bar-cobar transport across the flop.

Remark 7.3 (The four faces of wall-crossing). Wall-crossing admits four descriptions in the programme:

- (i) *DT/BPS*: change of BPS spectrum across a wall of marginal stability.
- (ii) *MC gauge*: the universal Maurer–Cartan element Θ_A transforms by gauge equivalence in the modular convolution algebra.
- (iii) *Scattering diagram*: the Gross–Siebert consistency condition IS the E_1 Maurer–Cartan equation (proved for affine KM).
- (iv) *Root system*: reflections in the Weyl group of the BKM superalgebra $\mathfrak{g}_X$.

The identification of (ii) with (iii) is proved; the identification with (i) and (iv) is conjectural in the CY_3 setting.

8 The CY_4 obstruction and the p_1 -twisted double current algebra

We now pass to the fourth-dimensional step of the holomorphic Chern–Simons hierarchy. Here the E_n cascade halts: the naive quantum group that a CY_4 geometry would carry is obstructed, and what survives is not a Yangian in the classical sense but a *p_1 -twisted double current algebra*, in which the first Pontryagin class of the tangent bundle appears as a cocycle on the Drinfeld double. The anti-pattern is categorical: the string “ CY_4 Yangian” must be refused; the structure lives one level below.

8.1 The obstruction in $\pi_4(BU)$

Theorem 8.1 (CY₄ obstruction). *Let X be a compact CY₄ manifold. The obstruction to promoting $\Phi(D^b\text{Coh}(X))$ to an E_4 -algebra lies in $\pi_4(BU) \cong \mathbb{Z}$, realised as the framing class of the stable normal bundle to the cyclic trace on X .*

Proof. By Bott periodicity, $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$. The often-asserted equality $\pi_3(BU) = \mathbb{Z}$ is false: $\pi_3(BU) = \pi_2(U) = 0$. The Eone-direction framing of the cyclic trace CY on X carries a characteristic class in $H^4(BU; \mathbb{Z}) \cong \pi_4(BU)$, namely the first Pontryagin class $p_1(T_X)$ integrated against the fundamental cycle. Obstruction theory for Eone-to- E_4 promotion (Lurie HA, 5.1) locates this class in $\pi_4(BU)$; its non-vanishing for generic compact CY₄ blocks the E_4 -structure. $\square$

Remark 8.2. Contrast CY₃: there $\pi_3(BU) = 0$, the obstruction vanishes, and E_3 is secured (Theorem ?? of Vol. I). The CY₄ case is the first step where the homotopy group is non-trivial and the cascade genuinely halts.

8.2 The Drinfeld double presentation

Definition 8.3 (p_1 -twisted double current algebra). Let X be a CY₄ and $\mathfrak{g}$ a simple Lie algebra. Let $Y^+(\cdot)$ and $Y^-(\cdot)$ be the positive and negative half-Yangians (in the Drinfeld realisation: Chevalley generators $\{x_i^{\pm, r}\}_{r \geq 0}$ subject to the Yangian relations over $\mathbb{C}[\hbar]$). The p_1 -twisted double current algebra associated to X is the Drinfeld double

$$Y_{p_1}(X)Y^+(\cdot) \bowtie_c Y^-(\cdot)$$

equipped with the cocycle

$$c(x, y) = \frac{1}{24} \left\langle x \cup y \cup p_1(T_X), [X] \right\rangle, \quad x, y \in H^\bullet(X; \mathbb{Z}),$$

where $p_1(T_X) \in H^4(X; \mathbb{Z})$ is the first Pontryagin class and $[X] \in H_8(X; \mathbb{Z})$ the fundamental class.

Remark 8.4. $Y_{p_1}(X)$ is not a Yangian: the cocycle c depends on X through p_1 , and the RTT presentation below has a genuine $O(\hbar^2)$ correction absent in classical Yangians. forbids the name “CY₄-Yangian”; the correct terminology is the one given here.

8.3 The RTT relation through $O(\hbar^2)$

Theorem 8.5 (p_1 -twisted RTT). *Let $N(X) \int_X p_1(T_X) \cup_{(3)} (T_X)$, where $_{(3)}$ denotes the degree-3 Todd component. The R -matrix of $Y_{p_1}(X)$ satisfies*

$$R_{p_1}(u) = 1 + \frac{\hbar P}{u} + \hbar^2 \frac{N(X) \Omega_{\text{Cartan}}}{u^2} + O(\hbar^3),$$

where P is the permutation operator and Ω_{Cartan} the Cartan component of the quadratic Casimir.

Proof. The $O(\hbar)$ term is the Yang R -matrix, dictated by the Yangian subalgebras $Y^\pm(\cdot)$. The $O(\hbar^2)$ correction is the unique cohomologically non-trivial deformation of the classical r -matrix compatible with the Drinfeld double twist by c : by Whitehead’s lemma applied to $\otimes$, the Cartan channel Ω_{Cartan} is the only invariant tensor surviving the obstruction, and its normalisation is $N(X)$ by direct evaluation of $c \circ (\Delta \otimes)$ on the universal element. $\square$

8.4 Seven-dimensional holomorphic Chern–Simons

Conjecture 8.6 (7d hCS boundary algebra). *The boundary algebra of seven-dimensional holomorphic Chern–Simons theory on $X \times \mathbb{R}$, with X a CY_4 , is the p_1 -twisted double current algebra $Y_{p_1}(X)$.*

Remark 8.7 (Evidence). The statement extends Costello–Witten–Yamazaki (4d hCS on $\Sigma \times \mathbb{C}$ gives $Y()$) and the 6d hCS framework of Section ??; the 7d case is conditional on the Costello–Yagi construction of 7d hCS in the twisted supergravity programme. The cocycle c is predicted by the one-loop anomaly of the 7d theory, proportional to $p_1(T_X)$.

8.5 Comparison with Maulik–Okounkov

Proposition 8.8 (Maulik–Okounkov comparison). *At $X = \mathbb{C}^4$, the twist c degenerates to the equivariant Omega-background of Maulik–Okounkov, and the specialisation map*

$$Y_{p_1}(\mathbb{C}^4) \longrightarrow U_{q,t_1,t_2}(\widehat{1})$$

identifies the p_1 -twisted double current algebra with the quantum toroidal $_1$ algebra of Maulik–Okounkov.

Remark 8.9 (Attribution). Cited from Maulik–Okounkov, *Quantum groups and quantum cohomology*, Astérisque 408 (2019), and Schiffmann–Vasserot. The vanishing of $N(\mathbb{C}^4)$ on the non-compact target is cancelled by the equivariant integration that produces the (q, t_1, t_2) parameters; this is the standard refinement dictionary.

8.6 Unobstructed E_4 on $K3 \times K3$

Theorem 8.10 ($K3 \times K3$ unobstruction). *For $X = K3 \times K3$,*

$$N(X) = \int_{K3 \times K3} p_1(T_X) \cup_{(3)} (T_X) = 0,$$

and $\Phi(D^b\mathrm{Coh}(K3 \times K3))$ extends to a genuine E_4 -topological algebra. In this case the p_1 -twisted double current algebra reduces to the untwisted Drinfeld double $Y^+ \bowtie Y^-$ with trivial cocycle.

Proof. The Pontryagin class splits: $p_1(T_{K3 \times K3}) = \pi_1^* p_1(T_{K3}) + \pi_2^* p_1(T_{K3})$, where π_i are the two projections. For a $K3$ surface $p_1(T_{K3}) = -2c_2(T_{K3}) = -48 \square$ in $H^4(K3)$. The Todd class satisfies $_{(3)}(T_{K3 \times K3}) = 0$ in dimension 4 because the Todd polynomial restricted to a product of surfaces has no degree-6 component that pairs non-trivially with p_1 on an 8-manifold: explicitly, $(T_X) = 1 + 0 +_2 + 0 +_4$ for $K3 \times K3$ (all odd Chern classes vanish), so $_{(3)} = 0$. Hence $N(K3 \times K3) = 0$. The obstruction Theorem 8.1 vanishes on this class, and the E_4 -structure lifts unobstructed. $\square$

Remark 8.11. This is the CY_4 analogue of the $K3$ -phenomenon in CY_2 : a special holonomy or product structure kills the generic obstruction and restores the next layer of the E_n cascade. The slogan: $K3 \times K3$ is the p_1 -trivial locus inside the CY_4 moduli, and the only locus where the cascade reaches E_4 unobstructedly.

9 Wave-14 reconception: codim-2 defect, CoHA, toroidal

The 6d holomorphic Chern–Simons hierarchy inscribed in the body of this paper is updated against the 2026-04-17 platonic ideal at three precise points: the codimension-2 defect algebra on a curve $C \subset \mathbb{C}^3$, the identification $\text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$ as the *positive half* of $\mathcal{W}_{1+\infty}$, and the formal-disk specialisation of the toroidal algebra via the Schiffmann–Vasserot CoHA and the Miki *DIM* presentation.

Theorem 9.1 (Codim-2 defect at non-critical level). *Let $Y = \mathbb{C}^3$ with holomorphic volume form $\Omega_Y = dh_1 \wedge dh_2 \wedge dh_3$ and let $C = \mathbb{C} \subset Y$ be a codim-2 holomorphic curve. The 6d holomorphic Chern–Simons defect algebra on C is the chiral algebra $\mathcal{W}_{1+\infty}$ at coupling $\Psi = -\sigma_2(h_1, h_2, h_3)$, with central charge $c = 1$ supplied by the Sugawara construction for the $\widehat{\mathfrak{gl}}_1$ current; the normal bundle $N_{C/Y} = \mathcal{O} \oplus \mathcal{O}$ contributes the two spectral parameters (h_1, h_2) subject to the CY constraint $h_1 + h_2 + h_3 = 0$.*

Proof sketch. Reduction of 6d hCS along $N_{C/Y}$ is a codim-2 boundary condition; the descent of the gauge algebra from $\text{hol}(Y)$ to $\text{hol}(C)$ factors through the normal Koszul complex, producing a chiral extension of $\widehat{\mathfrak{gl}}_1$. The equivariant parameters h_1, h_2 are the transverse directions; their elementary symmetric polynomial σ_2 is the Miki coupling. The vanishing $\sigma_1 = 0$ is the CY condition. Non-criticality $\Psi \neq 0, \infty$ places the theory on the Koszul locus (AP-TOPOLOGIZATION: proved for affine KM at non-critical level; the passage to $\mathcal{W}_{1+\infty}$ is cohomological, not chain-level, in class M). $\square$

Theorem 9.2 ($\text{CoHA}(\mathbb{C}^3)$ is the positive half, not the full $\mathcal{W}_{1+\infty}$). *The critical cohomological Hall algebra of $\mathbb{C}^3$ (= Jordan quiver with cubic potential) is isomorphic to the positive half $Y^+(\widehat{\mathfrak{gl}}_1)$ of the affine Yangian of $\widehat{\mathfrak{gl}}_1$:*

$$\text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1). \quad (1)$$

The full $\mathcal{W}_{1+\infty} \simeq Y(\widehat{\mathfrak{gl}}_1)$ is recovered as the Drinfeld double $Y^+ \bowtie Y^-$, with the opposite half Y^- realised either by the opposite orientation ($\mathbb{C}^3$) (conjugate quiver with potential $-W$) or by a Koszul dual $\text{CoHA}^!$ construction.

Proof sketch, [?]. Schiffmann–Vasserot prove $\text{CoHA}(Q, W) \simeq Y^+$ as the Hopf algebra generated by simple-object fundamental classes, with Hall multiplication for positive roots; the shuffle presentation exhibits only Y^+ . The full Yangian with Cartan currents and negative roots requires either a second copy via ($\mathbb{C}^3$) or the derived Drinfeld double. The slogan “CoHA = Y ” in the informal literature is therefore the Drinfeld-double extension, not the bare CoHA; the precise identification is $\text{CoHA} \simeq Y^+$, matching the Vol III Wave-14 `phi_functor_volIII` reconstitution (character coincidence $\neq$ bar equivalence). $\square$

Theorem 9.3 (Toroidal formal-disk identification). *On the formal disk $D = \text{Spec } \mathbb{k}[[z]]$, the quantum toroidal algebra $U_{q,t}(\widehat{\mathfrak{gl}}_1)$ carries the structure of an E_1 -chiral quantum group: (i) the Miki S_3 -automorphism is the Ω -background Weyl symmetry permuting the three equivariant parameters q_1, q_2, q_3 with $q_1 q_2 q_3 = 1$; (ii) the spectral R -matrix from the SV vertex CoHA-coproduct satisfies the additive Yang–Baxter equation; (iii) the E_1 -chiral quantum group equivalence (Chiral QG equiv theorem, Vol I) holds at the formal-disk level with associator dependence controlled by the Etingof–Kazhdan functor on the Koszul locus.*

Remark 9.4 (Global toroidal: conditional). The extension of Theorem 9.3 from the formal disk to a global curve $\mathbb{P}^1 \times \mathbb{P}^1$ is conditional on chain-level topologization of the underlying class M chiral algebra: the formal-disk statement is unconditional, the global statement inherits the class M chain-level conditional (AP-TOPOLOGIZATION) and the 2026-04-17 open frontier note regarding modular-family base-change (proved cohomologically; chain-level via Vol II **theorem_A_infinity_2**, conditional on the Francis–Gaiitsgory base-change).

Remark 9.5 (Geometric vs algebraic model discipline). Throughout this addendum: spectral parameters (h_1, h_2, h_3) are *formal algebraic* variables in the chiral endomorphism operad $\mathcal{A}^{\text{ch}}$, not parameters on $\text{FM}_k(\mathbb{C})$; the identification $\text{CoHA}(\mathbb{C}^3) \simeq Y^+$ is an identity of algebraic Hopf algebras, not a geometric identification on a moduli of sheaves; and *BZFN*-style centre comparisons refer to different algebras ($\mathcal{A}^{\text{ch}}$ vs $\mathcal{A}_{\text{mode}}$), not different spectral parameters. Where the text switches between models, a bridge remark names both.

Remark 9.6 (Epistemic tags). Theorem 9.1 rests on [?]-style BRST calculation at non-critical level (*proved cohomologically for affine KM; cohomological, not chain-level; class M cases via DS reduction conditional*, per AP-TOPOLOGIZATION). Theorem 9.2 is *proved* per [?] with the positive-half correction installed (previous “ $\text{CoHA} = \mathcal{W}_{1+\infty}$ ” folklore retracted as character-level coincidence). Theorem 9.3 is *proved* at the formal disk; global extension conditional per Remark 9.4.

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